Huey's Home Lab Build Documentation

Phase 2 — Active Directory, Domain Join & Network Configuration

Session Dates: May 31 – June 2, 2026

1. Phase 2 Overview

This document covers Phase 2 of the cybersecurity home lab build. Starting from a working GNS3 topology with Kali Linux and Windows Server 2025 connected, this phase covers fixing the VirtualBox UUID conflict, configuring Active Directory, setting up a domain controller, and joining Kali Linux to the hueylab.local domain.

2. Starting Point (End of Phase 1)

- GNS3 VM running in VMware Workstation Pro
- Huey's Enterprise (Kali Linux) connected to Windows Server 2025 in GNS3 project: Huey's Starship
- Static IPs configured: Kali 192.168.99.20 / Windows Server 192.168.99.10
- Both machines on the Inside network (192.168.99.0/24)
- Ping connectivity confirmed at 0% packet loss

3. Active Directory Configuration

3.1 Installation

- Removed AD FS (Active Directory Federation Services) which was accidentally installed first
- Installed AD DS (Active Directory Domain Services) via Server Manager > Add Roles and Features
- Set Administrator account password to Adman123# to meet AD DS prerequisites
- Promoted Windows Server to domain controller with new forest: hueylab.local
- DSRM (Directory Services Restore Mode) password set during promotion
- Server rebooted automatically after promotion

3.2 Domain Details

Setting	Value
Domain Name	hueylab.local
Forest	hueylab.local

Setting	Value
Domain Controller	WIN-430KUCVRBB4
Domain Controller IP	192.168.99.10
Administrator Password	Adman123#
DSRM Password	Set during promotion

3.3 Organizational Units

OU Name	Accidental Deletion Protection	Contents
The Sayian Race	Enabled	Kakarot Sayian, Vegeta Sayian, Gogeta group

3.4 Users Created

Full Name	Logon Name	Password	OU
Kakarot Sayian	ksayian	Password123!	The Sayian Race
Vegeta Sayian	vsayian	Password123!	The Sayian Race

3.5 Groups Created

Group Name	Type	Scope	Members	OU
Gogeta	Security	Global	ksayian	The Sayian Race

4. Kali Linux Domain Join

4.1 DNS Configuration

- Edited /etc/resolv.conf to point to Windows Server as primary DNS
- Added Google DNS as fallback: nameserver 8.8.8.8
- Locked resolv.conf with: sudo chattr +i /etc/resolv.conf
- Verified hueylab.local resolves to 192.168.99.10 with 0% packet loss

4.2 Package Installation

Since the Inside network has no internet access, Kali's Adapter 1 was temporarily switched from Generic Driver (UDPTunnel) to NAT to download packages. The following steps were taken:

- Switched VirtualBox Adapter 1 to NAT and ran sudo dhclient eth0 to get internet
- Updated Kali package signing key: downloaded from archive.kali.org and installed via gpg
- Ran sudo apt update to refresh package lists
- Installed required packages: realmd adcli krb5-user packagekit samba-common samba-common-bin
- Configured Kerberos realm as HUEYLAB.LOCAL with KDC at 192.168.99.10
- Switched Adapter 1 back to Generic Driver (UDPTunnel) after installation

4.3 Domain Join Command

Command used to join the domain:

```
sudo adcli join hueylab.local -U Administrator
```

Password used: Adman123#

- Domain join confirmed by KALI computer account appearing in Active Directory Users and Computers > Computers container

5. Errors & Troubleshooting Log

Error 1: VirtualBox UUID Conflict

Problem: After renaming Zee's Comp files to Huey's Enterprise, VirtualBox showed the VDI as Inaccessible. Selecting the renamed VDI in storage settings produced a UUID conflict error.

Solution: Ran the following command in CMD to assign a new UUID to the VDI:

```
"C:\Program Files\Oracle\VirtualBox\VBoxManage.exe" internalcommands
sethduuid "C:\Users\East39th\VirtualBox VMs\Planets\Zee_s Comp\Huey's
Enterprise.vdi"
```

Then repointed the storage in VirtualBox Settings > Storage to the renamed VDI. Also renamed the parent folder from Zee_s Comp to Huey's Enterprise in File Explorer.

Error 2: AD DS vs AD FS Confusion

Problem: Active Directory Federation Services (AD FS) was accidentally installed instead of Active Directory Domain Services (AD DS).

Solution: Completed the AD DS installation separately, then removed AD FS via Server Manager > Remove Roles and Features to keep the environment clean.

Error 3: Administrator Password Not Meeting Requirements

Problem: The AD DS prerequisites check failed with error: 'The local Administrator account becomes the domain Administrator account. The new domain cannot be created because the local Administrator account password does not meet requirements.'

Solution: Opened CMD as Administrator on Windows Server and ran: net user Administrator Adman123# to set a compliant password. Then clicked Rerun prerequisites check.

Error 4: sssd Packages Not Available in Kali

Problem: sudo realm join --user=Administrator hueylab.local failed because sssd, sssd-tools, libnss-sss, and libpam-sss are no longer available in the current Kali rolling repositories.

Solution: Used adcli directly to join the domain: sudo adcli join hueylab.local -U Administrator. This bypasses the sssd requirement and communicates directly with Active Directory.

Error 5: Kali Package Signing Key Error

Problem: sudo apt update failed with: Missing key 827C8569F2518CC677FECA1AED65462EC8D5E4C5, which is needed to verify signature.

Solution: Downloaded and installed the new Kali signing key:

```
sudo wget -O /tmp/kali-key.asc https://archive.kali.org/archive-key.asc
sudo gpg --dearmor -o /usr/share/keyrings/kali-archive-keyring.gpg /tmp/kali-key.asc
```

Error 6: Kali No Internet Access on Inside Network

Problem: The Inside network (192.168.99.0/24) is completely isolated with no internet route. Kali could not download packages needed for domain join.

Solution: Temporarily switched VirtualBox Adapter 1 from Generic Driver (UDPTunnel) to NAT, ran sudo dhclient eth0 to get a DHCP address on the NAT network (10.0.2.x), downloaded all required packages, then switched back to Generic Driver after installation.

Error 7: Kerberos Preauthentication Failed

Problem: sudo adcli join hueylab.local -U Administrator returned: Couldn't authenticate as: Administrator@HUEYLAB.LOCAL: Preauthentication failed.

Solution: The password being entered was incorrect. Reset the Administrator password on Windows Server with: net user Administrator Adman123# and used that exact password in the adcli command.

6. Final Network Configuration

Device	IP Address	Subnet	Network	Role
Huey's Enterprise (Kali)	192.168.99.20	255.255.255.0	Inside	Attack Machine / Domain Member
Windows Server 2025	192.168.99.10	255.255.255.0	Inside	Domain Controller
GNS3 VM	192.168.108.128	255.255.255.0	VMnet1	GNS3 Backend

7. Connectivity Verification

Test	Command	Result
Kali to Windows Server (IP)	ping 192.168.99.10	0% packet loss
Kali to Domain Name	ping hueylab.local	Resolves to 192.168.99.10, 0% packet loss
Domain Join Verification	Active Directory > Computers	KALI computer account visible

8. Session Startup Procedure

- Start GNS3 VM manually in VMware Workstation
- Open GNS3 Windows application
- Open the Huey's Starship project
- Click the green Play button in GNS3
- If Kali doesn't start automatically, power it on manually in VirtualBox
- Verify eth0 on Kali shows 192.168.99.20 (run: ip addr show eth0)
- Verify Windows Server shows 192.168.99.10 (run: ipconfig)
- Run ping test to confirm connectivity: ping 192.168.99.10
- Verify hueylab.local resolves: ping hueylab.local

9. Current Lab Status

Component	Status	Details
VMware Workstation Pro	Installed	Version 26H1
GNS3 + GNS3 VM	Running	Version 2.2.59, IP: 192.168.108.128
Huey's Enterprise (Kali)	Domain Joined	192.168.99.20 / hueylab.local

Component	Status	Details
Windows Server 2025	Domain Controller	192.168.99.10 / WIN-430KUCVRBB4.hueylab.local
Active Directory	Configured	Domain: hueylab.local, OU: The Sayian Race
Users	Created	ksayian (Kakarot), vsayian (Vegeta)
Groups	Created	Gogeta (members: ksayian)
GNS3 Project	Active	Huey's Starship

10. Next Steps

- Configure DHCP role on Windows Server 2025
- Practice Kerberoasting attacks from Kali against hueylab.local
- Set up BloodHound for Active Directory enumeration
- Practice Pass-the-Hash attacks and monitor Windows Event Logs
- Add a router/switch between VMs in GNS3 for more realistic topology
- Add vsayian (Vegeta) to a security group
- Install VMware VIX API to enable full GNS3 control of VMware VMs
- Document attack and defense techniques as they are practiced
- Consider adding a Windows 10 workstation VM as a third machine